

JMP Exam Notes

ONE MEAN • POWER • TWO PROPORTIONS

➤ = then click · ▼ = red triangle on that report's title bar · ➡ = where the answer appears · reject H_0 when $p \leq \alpha$

SETTING UP

Open the data. File ➤ Open ➤ the .jmp file. A quantitative column must read **Continuous** — right-click the column header ➤ Column Info ➤ Modeling Type.

Categorical or quantitative? A number you could average is quantitative; a label sorting units into groups is categorical. Quantitative → a mean; categorical → a proportion.

Hypotheses. $H_0: \mu = \mu_0$ always. "less / decreased" → $H_1: \mu < \mu_0$ · "greater / increased" → $>$ · "different / changed" → $\neq$. Never put $\bar{x}$ in a hypothesis.

Shape, $\bar{x}$, s, n. Analyze ➤ Distribution ➤ Y, Columns = variable ➤ OK ➤ histogram for shape; **Summary Statistics** for Mean, Std Dev, N, Std Err Mean.

TESTING ONE MEAN

t test (σ not given). Analyze ➤ Distribution ➤ Y = variable ➤ OK ➤ ▼variable ➤ Test Mean ➤ type μ_0 ➤ leave the SD box empty ➤ OK ➔ **Test Mean** panel, **t Test** row.

z test (σ given). Same path, but also fill "Enter True Standard Deviation to do z-test rather than t test" → the panel now shows a **z Test** row.

Pick the right p-value. H_1 is $\neq$ → **Prob>|t|** · H_1 is $>$ → **Prob>t** · H_1 is $<$ → **Prob<t**. Ignore the Signed-Rank row.

No data, only summary stats. Distribution needs raw data, so compute by hand: $t = (\bar{x} - \mu_0) / (s/\sqrt{n})$, $df = n - 1$. If they give s^2 , take the square root first.

CONDITIONS AND INTERVALS

Conditions for the t test. Independence: random sample, under 10% of the population, one measurement per unit. **Normality:** stated bell-shaped, **or** $n \geq 30$ (CLT), **or** symmetric histogram with no outliers.

Check normality in JMP. ▼variable ➤ Continuous Fit ➤ Normal ➤ ▼Fitted Normal ➤ Goodness of Fit ➔ **Shapiro-Wilk Prob<W**; above α means normality is reasonable. Add ▼variable ➤ Normal Quantile Plot and say the points follow the line inside the bands.

Confidence interval for μ . **Summary Statistics** already lists Upper / Lower 95% Mean. Other levels: ▼variable ➤ Confidence Interval ➤ 0.90 / 0.95 / 0.99. By hand $\bar{x} \pm t(\alpha/2, n-1) \cdot s/\sqrt{n}$.

Probability about an average. $z = (\bar{x} - \mu) / (\sigma/\sqrt{n})$, or Cols ➤ New Column ➤ Formula ➤ Probability ➤ Normal Distribution(a, μ , $\sigma/\sqrt{n}$) which gives $P(\bar{x} < a)$; subtract from 1 for "more than".

SAMPLING DISTRIBUTION & POWER

Describe the sampling distribution of $\bar{x}$ when H_0 is true. Give all three. **Shape:** bell-shaped / normal — because the population is normal, or by the CLT when $n \geq 30$. **Mean:** μ_0 . **Standard deviation:** $\sigma/\sqrt{n}$ (this is the standard error).

What values of $\bar{x}$ lead to rejection? critical value = $\mu_0 + (z)(SE)$, with $SE = \sigma/\sqrt{n}$ and z signed by the tail: left -1.645 , right $+1.645$ at $\alpha = .05$; two-tailed $\mu_0 \pm 1.96 \cdot SE$. Reject when $\bar{x}$ is past it.

$\beta = P(\text{Type II error})$. Re-standardize that same critical value using the **true** mean: $z = (\bar{x}c - \mu_{\text{true}}) / (\sigma/\sqrt{n})$, then $\beta = P(\bar{x} \text{ lands in the fail-to-reject region})$.

Power = $1 - \beta$. "Probability of detecting this difference", "probability of rejecting H_0 when the true mean is ____" and "probability of correctly concluding" all mean power. "Probability of failing to reject" and "incorrectly failing to reject" mean β .

Worked example. $\mu_0 = 50$, $\sigma = 21$, $n = 36$, $\alpha = .05$, left-tailed. $SE = 21/6 = 3.5$; $\bar{x}c = 50 - 1.645(3.5) = \mathbf{44.24}$. If $\mu_{\text{true}} = 43$: $z = (44.24 - 43)/3.5 = .354$, $\beta = \mathbf{.3613}$, power = **.6387**.

Power from JMP. DOE ➤ Design Diagnostics ➤ Sample Size and Power ➤ One Sample Mean, fill Alpha, Std Dev, Difference to Detect and Sample Size, leave **Power** blank ➤ Continue. Leaving **Sample Size** blank instead answers "how big a sample do I need". It is two-sided by default, so trust the hand computation on one-tailed problems.

Sketch. Two normal curves on one axis: top at μ_0 with the tail past $\bar{x}c$ shaded = α ; bottom at μ_{true} with the fail-to-reject side = β , the rest = power. One vertical line at $\bar{x}c$ through both.

TWO PROPORTIONS — INDEPENDENT GROUPS

Type in a 2x2 count table. File ➤ New ➤ Data Table with three columns — Group (Nominal), Outcome (Nominal), Count (Continuous) — and one row per cell, e.g. (Help, Accepted, 222).

Analyze it. Analyze ➤ Fit Y by X ➤ Y = Outcome ➤ X = Group ➤ Freq = Count ➤ OK ➔ **Contingency Table** and **Tests**. Forgetting the **Freq** box makes JMP analyze only 4 observations.

Point estimate. ▼Contingency Table ➤ Row %. Each $\hat{p}$ is the cell divided by **its own row total**, never the grand total; the estimate is $\hat{p}_1 - \hat{p}_2$. Example: $222/515 - 55/190 = .4311 - .2895 = \mathbf{.1416}$.

Hypotheses. $H_0: p_1 - p_2 = 0$. If the rates are the same in both groups the difference is zero. "improves" → $>$ · "reduces" → $<$ · "different" → $\neq$ · 0.

Assumptions. Independent random samples or random assignment; one response per person; all four cell counts ≥ 5 (exactly $n_1\hat{p}_1$, $n_1\hat{q}_1$, $n_2\hat{p}_2$, $n_2\hat{q}_2 \geq 5$).

Test and interval. ▼Contingency Analysis ➤ Two Sample Test for Proportions ➔ the difference, its standard error, **Prob>|z|** / **Prob>z** / **Prob<z**, and the confidence interval. **Pearson ChiSquare** in the Tests panel is the same two-sided test ($\chi^2 = z^2$).

By hand. pooled $\hat{p} = (x_1+x_2)/(n_1+n_2)$; $z = (\hat{p}_1-\hat{p}_2) / \sqrt{[\hat{p}(1-\hat{p})(1/n_1+1/n_2)]}$. The interval is **not** pooled: $(\hat{p}_1-\hat{p}_2) \pm z_{\alpha/2}\sqrt{[\hat{p}_1\hat{q}_1/n_1 + \hat{p}_2\hat{q}_2/n_2]}$.

TWO PROPORTIONS — PAIRED

Label the table. **a** and **d** are the ties (no change); **b** and **c** are the discordant pairs that changed in each direction. Smoking: $a = 20$, $b = 5$, $c = 16$, $d = 9$, $n = 50$.

Ties, n', T. ties = $a + d$, $n' = b + c$, and T is the count in the direction H_1 predicts. Smoking: ties = 29, $n' = 21$, $T = 16$. $H_0: p_b = p_c$, i.e. changes are equally likely both ways.

p-value, exact binomial. Under H_0 , $T \sim \text{Binomial}(n', 0.5)$. Cols ➤ New Column ➤ Formula ➤ Discrete Probability ➤ Binomial Distribution(.5, n' , k) gives $P(T \leq k)$. Smoking: $1 - \text{BinDist}(5, 21, 15) = \mathbf{.0133} < .05$, so reject — the video was effective.

Or from JMP. Analyze ➤ Fit Y by X ➤ Y = After, X = Before, Freq = Count ➤ OK ➤ ▼ ➤ Agreement Statistic ➔ the **test of symmetry** is McNemar's test, but two-sided only.

Interval. $(b-c)/n \pm z_{\alpha/2}\sqrt{[(a+d)(b+c) + 4bc] / n^3}$. Smoking gives $(- .389, -.051)$, entirely below 0. Puzzle gives $(- .196, .077)$, which contains 0 — no significant difference.

FORMULAS & CRITICAL VALUES

$S_{xx} = \sum (x - \bar{x})^2$ $S_{yy} = \sum (y - \bar{y})^2$ $S_{xy} = \sum (x - \bar{x})(y - \bar{y})$ · $b_1 = S_{xy}/S_{xx}$ $b_0 = \bar{y} - b_1\bar{x}$ $r = S_{xy}/\sqrt{(S_{xx} \cdot S_{yy})}$

df: $n - 1$ for one mean · $n - 2$ for a simple-regression slope · $n - k - 1$ for a multiple-regression coefficient · k and $n - k - 1$ for the overall F.

z critical values: one-tailed $\alpha = .10 \rightarrow 1.282$, $.05 \rightarrow 1.645$, $.025 \rightarrow 1.960$, $.01 \rightarrow 2.326$, $.005 \rightarrow 2.576$. Two-tailed 90% → 1.645, 95% → 1.960, 98% → 2.326, 99% → 2.576.

JMP formula functions: t Quantile($1-\alpha/2$, df), Normal Quantile($1-\alpha$), t Distribution(t, df) (left tail), $2*(1 - t \text{ Distribution}(|t|, df))$ for a two-sided p-value.

FILL-IN-THE-BLANK ANSWERS

DEFINE THE PARAMETER — μ = the true mean _____ (with units) for all _____ .

INTERPRET THE TEST STATISTIC — The sample mean is _____ standard errors [above / below] _____ , the value H_0 claims.

INTERPRET THE P-VALUE — If the true mean really were _____ , there is a _____ % chance of getting a sample mean at least this far from _____ by random sampling alone.

CONCLUSION — WORKS FOR EVERY TEST — Since $p =$ _____ [$<$ / $>$] $\alpha =$ _____ , we [reject / fail to reject] H_0 . There [is / is not] sufficient evidence to conclude that _____ .

Last blank → **ONE MEAN:** — the mean [variable] is [less than / greater than / different from] ____ · **TWO PROPORTIONS:** — the proportion of ____ is [higher / lower / different] for ____ than for ____ · **SLOPE:** — [x] is a useful predictor of [y] · **OVERALL F:** — the model is useful for predicting [y] · **QUADRATIC:** — the quadratic term improves the model. Never write "accept H_0 ."

INTERPRET ANY CONFIDENCE INTERVAL — We are _____ % confident that _____ is between _____ and _____ .

Middle blank → **M:** — the true mean [variable] for all [population] · **P1–P2:** — the difference in the proportion of ____ between ____ and ____ · **B1:** — the average change in [y] for each additional one [x-unit] · **MEAN RESPONSE:** — the mean [y] for all [units] with $x =$ ____ · **PREDICTION:** — a single [unit] with $x =$ ____ will have [y]. If the interval contains 0 → no significant difference / not a useful predictor.

TYPE I ERROR IN CONTEXT — We conclude that _____ when in reality _____ . The probability of this is $\alpha =$ _____ .

TYPE II ERROR IN CONTEXT — We conclude there is not enough evidence that _____ when in reality _____ is true. The probability of this is $\beta =$ _____ .

DESCRIBE THE SAMPLING DISTRIBUTION — **Shape:** approximately normal, because _____ . **Mean:** $\mu =$ _____ . **Standard deviation:** $\sigma/\sqrt{n} =$ _____ / $\sqrt{_____} =$ _____ .

Reason blank: "the population is stated to be normal" or " $n \geq 30$, so the CLT applies".

CHECK THE T-TEST CONDITIONS — **Independence:** the _____ are a random sample and fewer than 10% of all _____ . **Normality:** _____ .

POWER AND B — If the true mean is really _____ , there is a _____ % chance we [correctly reject / fail to reject] H_0 . "Correctly reject" = power = $1 - \beta$. "Fail to reject" = $\beta = P(\text{Type II error})$.

WHICH ERROR WAS POSSIBLE — We [rejected / failed to reject] H_0 , so if an error was made it must be a Type [I / II] error: _____ .

LARGE-SAMPLE CONDITION, TWO PROPORTIONS — The counts _____ , _____ , _____ and _____ are all at least 5, so both samples are large enough.

JMP Exam Notes

CORRELATION & THE FITTED LINE

Which variable is which. "Predict A using B" → y = A goes in **Y, Response**, x = B goes in **X, Factor**.

Scatterplot. Analyze > Fit Y by X > Y, Response = response > X, Factor = factor > OK.

Add the line. ▾Bivariate Fit > Fit Line → the equation is printed as text just above **Summary of Fit**.

Correlation r. Analyze > Multivariate Methods > Multivariate > both variables in Y, Columns > OK → **Correlations**, the off-diagonal value. Or r = $\pm\sqrt{\text{RSquare}}$ taking the sign of the slope. $|r| \geq .8$ strong, $.5$ -.8 moderate, below $.5$ weak.

Equation of the least squares line. Analyze > Fit Model > Y = response > select x > Add > Run → **Parameter Estimates**, **Estimate** column: Intercept row = b_0 , x row = b_1 .

Predict. Substitute into $\hat{y} = b_0 + b_1x$. Watch units — x in thousands means 34,000 is x = 34. Don't extrapolate past the data.

CHECKING THE CONDITIONS

Save what you need. ▾Response y > Save Columns > Residuals, then repeat for **Predicted Values** and **Studentized Residuals** → new columns at the right of the data table.

Normality of the residuals. Analyze > Distribution > Y = Residual > OK > ▾Residual > Normal Quantile Plot, then Continuous Fit > Normal > ▾Fitted Normal > Goodness of Fit → **Shapiro-Wilk**.

Residual plot. ▾Response y > Row Diagnostics > Plot Residual by Predicted. Random band = fine · arch or U = wrong form · funnel = variance not constant · studentized past ± 3 = outlier.

IS THE PREDICTOR USEFUL?

Test the slope. Fit Model > Run > Parameter Estimates, on the x row → **t Ratio** is the statistic, **Prob>|t|** the two-sided p-value, **Std Error** is $SE(b_1)$. $H_0: \beta_1 = 0$, $df = n - 2$. State H_0 , H_1 , t, p and a conclusion in context.

One-sided? Halve **Prob>|t|**, but only if the sign of b_1 agrees with H_1 ; if it disagrees the p-value is above $.5$ and you fail to reject.

Testing a slope against a value other than 0. JMP only tests $\beta_1 = 0$. Either check whether the CI contains that value, or compute $t = (b_1 - c)/SE(b_1)$, $df = n - 2$.

Variation explained. Summary of Fit → RSquare. Say "___% of the variation in [y] is explained by the linear relationship with [x]."

THE THREE INTERVALS

CI for the slope. Right-click inside **Parameter Estimates** > Columns > Lower 95% + Upper 95% → read them on the **x** row, not the Intercept row. If it excludes 0, x is a useful predictor.

99% or 98% instead. ▾Response y > Set Alpha Level > 0.01 (or **Other** → 0.02), then re-read the columns. If locked at 95%: $b_1 \pm t(\alpha/2, n-2) \cdot \text{Std Error}$, with t from t Quantile($1-\alpha/2$, $n-2$).

CI for the mean response — "the average y for ALL cases with x = ____". Add a data row with that x and **Y left blank** (a blank Y keeps it out of the fit), then Fit Model > Run > ▾Response y > Save Columns > Mean Confidence Interval → **Lower / Upper 95% Mean** on that row.

Prediction interval — "predict for ONE new case". Same setup, then Save Columns > Indiv Confidence Interval → **Lower / Upper 95% Indiv**, always wider. Singular wording ("a truck", "a school") means Indiv.

Bands on the plot. ▾Linear Fit > Confid Curves Fit or Confid Curves Indiv — graph only.

Unusual points. Click the point to highlight its row, then Rows > Exclude/Unexclude and re-Run. "Votes more than predicted" = actual — predicted = that row's residual.

MULTIPLE REGRESSION

Screen the predictors first. Analyze > Multivariate Methods > Multivariate > response + all predictors in Y, Columns > OK → **Correlations** and a **Scatterplot Matrix**. Predictor-vs-predictor $|r|$ above $\sim .8$ signals multicollinearity.

Fit it. Analyze > Fit Model > Y = response > highlight all predictors > Add > Run → **Parameter Estimates** gives $\hat{y} = b_0 + b_1x_1 + \dots + b_kx_k$.

Is the model useful? Analysis of Variance → F Ratio and Prob > F. H_0 : all $\beta = 0$; H_1 : at least one $\beta \neq 0$. Rejecting means at least one predictor is useful, not all of them.

Is one predictor useful, adjusting for the rest? Parameter Estimates, that predictor's row → **t Ratio**, **Prob>|t|**, $df = n - k - 1$. The **Effect Tests** table gives the same p-value as an F. A predictor can look strong alone and still be non-significant here.

CI for a coefficient. Right-click **Parameter Estimates** > Columns > Lower 95% + Upper 95%; set a different level first with **Set Alpha Level**. Keep "holding the others constant" in the interpretation.

Conditions. Save **Residuals** and **Studentized Residuals**; ▾Response y > Row Diagnostics > Plot Residual by Predicted; Distribution on the residuals for the NQ plot and Shapiro-Wilk. Multicollinearity: right-click **Parameter Estimates** > Columns > VIF — over 5 warning, over 10 a problem.

CHOOSING A MODEL

Compare with adjusted R². Summary of Fit → RSquare Adj, highest wins. Plain R² always rises when a predictor is added, so it cannot compare models of different sizes; $R^2_{\text{adj}} = 1 - (1 - R^2)(n-1)/(n-k-1)$ penalizes extras.

CORRELATION · SIMPLE, MULTIPLE & POLYNOMIAL REGRESSION

All possible models. Fit Model > add all > Personality = Stepwise > Run > ▾ > All Possible Models → every subset with RSquare, RSquare Adj, Cp, AICC, BIC.

Backward selection. Fit Model > add all > Stepwise > Run, set Direction = **Backward**, Stopping Rule = **P-value Threshold**, Prob to Leave = **0.1**, click **Go** → **Current Estimates**, a check under **Entered** = kept. Then **Make Model** → **Run**. By hand: drop the largest p-value above 0.1, refit, repeat — one at a time.

Predict from the chosen model. Plug into that model's equation, or add a row with those x's and a blank Y and save **Predicted Values**, or use ▾Response y > Factor Profiling > Profiler.

POLYNOMIAL (QUADRATIC) REGRESSION

Quick route. Analyze > Fit Y by X > Y, X > OK > ▾Bivariate Fit > Fit Polynomial > 2, quadratic → a **Polynomial Fit Degree=2** block with the uncentered equation printed above it.

Full route (use this for intervals). Analyze > Fit Model > Y > select x > Macros > Polynomial to Degree with Degree = 2 > Run → **Parameter Estimates** gains an **x*x** row.

Centering. JMP centers polynomial terms by default, printing $(x-\bar{x}) \cdot (x-\bar{x})$, so the coefficients are not the raw b_0, b_1, b_2 . Uncheck ▾Response y > Center Polynomials and Run again.

Is the quadratic useful? Test the squared term specifically: **Parameter Estimates**, the **x*x** row (or **Effect Tests**) → **Prob>|t|**. $H_0: \beta_2 = 0$ means a line is enough. The overall F can be significant just from the linear part, so don't use it here.

Change in y is no longer constant. change = $b_1(x_2 - x_1) + b_2(x_2^2 - x_1^2)$. From 0 to 1 that is $b_1 + b_2$; from 2 to 3 it is $b_1 + 5b_2$. The two differ — that is exactly the point of asking both.

Intervals from a quadratic. Add the row with x and a blank Y, fit with **Polynomial to Degree 2**, then **Save Columns** → **Mean Confidence Interval** for the average of all such cases, **Indiv Confidence Interval** for one specific new case.

FILL-IN-THE-BLANK ANSWERS — REGRESSION

COMMENT ON THE SCATTERPLOT —There is a **[positive / negative]**, **[weak / moderate / strong]**, **[linear / curved]** relationship between _____ and _____.

Add if needed: "There is an unusual point at ____." If it curves, say a quadratic model should be considered.

INTERPRET R — $r =$ _____ indicates a **[weak / moderate / strong]** **[positive / negative]** linear relationship between _____ and _____.

WRITE THE EQUATION —Predicted _____ = _____ + _____ (_____).

INTERPRET THE SLOPE —For each additional one _____, the predicted (average) _____ **[increases / decreases]** by _____.

INTERPRET THE INTERCEPT —When _____ is 0, the predicted _____ is _____.

Add the caveat when $x = 0$ is outside the data or impossible: "but $x = 0$ is outside the range of the data, so this is not meaningful."

INTERPRET R² — _____ % of the variation in _____ is explained by the **[linear / quadratic]** relationship with _____.

USEFUL PREDICTOR? —Since $p =$ _____ [$< / >$] $\alpha =$ _____, we **[reject / fail to reject]** $H_0: \beta_1 = 0$. _____ **[is / is not]** a useful predictor of _____.

CI FOR THE SLOPE —We are _____ % confident that for each additional one _____, the average _____ changes by between _____ and _____.

CI FOR THE MEAN RESPONSE —We are _____ % confident that the mean _____ for all _____ with _____ = _____ is between _____ and _____.

PREDICTION INTERVAL —We are _____ % confident that a single _____ with _____ = _____ will have a _____ between _____ and _____.

CONDITIONS FROM THE RESIDUAL PLOT —The residual plot shows _____, so the **[linearity / constant variance]** condition **[is / is not]** met. Shapiro-Wilk gives $p =$ _____ [$> / <$] α , so normality **[is / is not]** reasonable.

First blank: "a random band around zero" · "an arched pattern" · "a funnel shape".

INTERPRET A COEFFICIENT (MULTIPLE REGRESSION) —For each additional one _____, the predicted _____ **[increases / decreases]** by _____, **holding all other predictors constant**.

IS THE MODEL USEFUL? (OVERALL F) —Since $p =$ _____ [$< / >$] $\alpha =$ _____, we **[reject / fail to reject]** $H_0: \beta_1 = \beta_2 = \dots = 0$. The model **[is / is not]** useful for predicting _____.

ONE PREDICTOR, ADJUSTING FOR THE REST —Adjusting for the other predictors, _____ **[is / is not]** a useful predictor of _____ (t = _____, $p =$ _____).

CHOOSING A MODEL —The model with _____ has the highest adjusted R² (_____), so it is the preferred model.

IS THE QUADRATIC USEFUL? —The squared term has $p =$ _____ [$< / >$] $\alpha =$ _____, so the quadratic model **[is / is not]** useful for predicting _____.